

Detailed Summary

"Clinical Trial Biomarker Analytics and Advanced Data Visualization Pipelines" documents an extensive semester-long exploration of clinical trial biomarkers and advanced graphical design principles. The project transitions from basic programmatic charting to hand-drawn prototyping, ultimately culminating in highly dense, tailored infographics designed to maximize visual clarity and strategic biological insights.

Initial Analytical Framework & Core Datasets

The initial stages utilized trial spreadsheets (such as the "CTO sheet" and edited data charts) to establish a baseline vocabulary of clinical trial biomarkers and chart formats. Early automated charting examined how biomarker implementations scale across specific parameters:

- **Biomarker Categorization:** Analysis separated clinical data trackers into three main structural types: **Fluid**, **Imaging**, and **Digital** biomarkers.
- **Trial Phase Allocation:** Across clinical trial stages, Fluid biomarkers emerged as the absolute majority, whereas Imaging biomarkers demonstrated a steady volume increase in later phases (Phase 2 and Phase 3), and Digital indicators remained the least utilized.
- **Functional Use Distribution:** Programmatic bar charts tracking baseline totals surfaced that biomarkers are disproportionately utilized to track secondary outcomes (295 instances), followed by inclusion criteria (132 instances), alternative metrics (114 instances), and primary outcomes (79 instances).

Mid-Semester Prototyping (Hand-Drawn Sketches)

To circumvent the standard constraints of classical bar and pie charts, the author experimented with manual sketching to ideate unique radial and multi-tiered geometric spaces:

1. **The Concentric Arch Profile:** A blueprint utilizing color-coded ascending bars configured from the center outward to cleanly illustrate data frequency trends across multiple observation periods.
2. **The Tiered Proportional Ring:** A structural model where tier path widths dynamically scale to represent the relative volume distributions of Fluid, Imaging, and Digital assets.
3. **The Categorized Proximity Bubble:** An array grouping therapeutic classes into centralized clusters by biomarker type, mapping total instances via localized bubble size to maintain a crisp visual hierarchy.

Advanced Analytics & Multi-Dimensional Visualizations

The closing portfolio demonstrates the execution of multi-dimensional graphics mapping out cross-relational trial paths:

- **Alluvial Flow Mapping:** The team constructed comprehensive alluvial flow networks tracking how clinical trial targets flow smoothly into specific biomarker types, measurement sources, and

eventually into highly specified **CADRO** (Cognitive and Alzheimer's Disease Research Office) classifications.

- **Co-Occurrence Matrix Analysis:** A series of dense heatmaps evaluated localized category clustering. The intensity matrices proved that Secondary Outcomes and Inclusion Criteria anchor intensely with CADRO categories A (Amyloid) and B (Tau). Furthermore, cross-tabulations verified that Fluid options map tightly with CSF (Cerebrospinal Fluid) and Blood filters, while Imaging elements anchor directly with category R (Risk factors).
- **The Fluid Pipeline Layout:** A custom voronoi/binning compartmental matrix was engineered to model fluid-based tracking networks. In this layout, the physical section size maps execution frequency, background color notes the biological fluid origin (Blood vs. CSF), and localized vector geometric anchors define core physiological paths (such as Amyloid, Tau, Neurodegeneration, or Vascular mechanisms).